

Mathematical Foundations of Machine Learning

Assignment 4 Solutions

Question 1 (15 points)

Consider $\mathbb{R}^3$ equipped with the inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle_G := \mathbf{x}^\top \mathbf{G} \mathbf{y}, \quad \mathbf{G} = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}(t) = \begin{bmatrix} t \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

1. (4 points) Find all $t \in \mathbb{R}$ such that $\mathbf{u} \perp \mathbf{v}(t)$ with respect to $\langle \cdot, \cdot \rangle_G$.
2. (3 points) For the value of t found in (1), determine whether $\mathbf{u} \perp \mathbf{v}(t)$ holds under the standard dot product on $\mathbb{R}^3$.
3. (8 points) Let $L = \text{span}\{\mathbf{u}\}$. Compute the orthogonal projection $\eta_L(\mathbf{w})$ with respect to $\langle \cdot, \cdot \rangle_G$ and the distance $d_G(\mathbf{w}, L)$.

Solution:

1. We need $\langle \mathbf{u}, \mathbf{v}(t) \rangle_G = 0$:

$$\langle \mathbf{u}, \mathbf{v}(t) \rangle_G = \mathbf{u}^\top \mathbf{G} \mathbf{v}(t) = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} t \\ 1 \\ -1 \end{bmatrix}.$$

First compute $\mathbf{G} \mathbf{v}(t) = \begin{bmatrix} 3t+1 \\ t+1 \\ -1 \end{bmatrix}$. Hence

$$\langle \mathbf{u}, \mathbf{v}(t) \rangle_G = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3t+1 \\ t+1 \\ -1 \end{bmatrix} = 3t.$$

Therefore $t = 0$.

2. For $t = 0$, $\mathbf{v}(0) = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$. Under the standard dot product,

$$\mathbf{u}^\top \mathbf{v}(0) = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = -1 \neq 0,$$

so they are not orthogonal under the standard dot product.

3. Write $\eta_L(\mathbf{w}) = \alpha \mathbf{u}$. The residual $\mathbf{w} - \alpha \mathbf{u}$ must satisfy

$$\langle \mathbf{w} - \alpha \mathbf{u}, \mathbf{u} \rangle_G = 0 \Rightarrow \alpha = \frac{\langle \mathbf{w}, \mathbf{u} \rangle_G}{\langle \mathbf{u}, \mathbf{u} \rangle_G}.$$

Compute $\langle \mathbf{w}, \mathbf{u} \rangle_G = \mathbf{w}^\top \mathbf{G} \mathbf{u}$. Since $\mathbf{G} \mathbf{u} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$, we get

$$\langle \mathbf{w}, \mathbf{u} \rangle_G = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} = 4.$$

Also $\langle \mathbf{u}, \mathbf{u} \rangle_G = \mathbf{u}^\top \mathbf{G} \mathbf{u} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} = 5$. Hence

$$\eta_L(\mathbf{w}) = \frac{4}{5} \mathbf{u} = \begin{bmatrix} 4/5 \\ 0 \\ 4/5 \end{bmatrix}.$$

Let $\mathbf{r} = \mathbf{w} - \eta_L(\mathbf{w}) = \begin{bmatrix} -4/5 \\ 1 \\ 1/5 \end{bmatrix}$. Then

$$d_G(\mathbf{w}, L) = \|\mathbf{r}\|_G = \sqrt{\mathbf{r}^\top \mathbf{G} \mathbf{r}}.$$

Compute $\mathbf{G} \mathbf{r} = \begin{bmatrix} 3(-4/5) + 1 \cdot 1 \\ 1(-4/5) + 2 \cdot 1 + 1(1/5) \\ 1 \cdot 1 + 2(1/5) \end{bmatrix} = \begin{bmatrix} -7/5 \\ 7/5 \\ 7/5 \end{bmatrix}$, hence

$$\mathbf{r}^\top \mathbf{G} \mathbf{r} = \begin{bmatrix} -4/5 & 1 & 1/5 \end{bmatrix} \begin{bmatrix} -7/5 \\ 7/5 \\ 7/5 \end{bmatrix} = \frac{28}{25} + \frac{7}{5} + \frac{7}{25} = \frac{14}{5}.$$

Therefore

$$d_G(\mathbf{w}, L) = \sqrt{\frac{14}{5}}.$$

Question 2 (15 points)

Let

$$\mathbf{Q} = \frac{1}{3} \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}.$$

1. (7 points) Prove that $\mathbf{Q}$ is an orthogonal matrix.
2. (8 points) Let $S = \{\mathbf{v} \in \mathbb{R}^3 : \mathbf{Q}\mathbf{v} = \mathbf{v}\}$. Find a basis for S . Then pick a nonzero $\mathbf{v}_0 \in S$ and prove that if $\mathbf{y} \perp \mathbf{v}_0$ for some $\mathbf{y} \in \mathbb{R}^3$, then $\mathbf{Q}\mathbf{y} \perp \mathbf{v}_0$.
Finally, for $\mathbf{y} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$, compute the angle between $\mathbf{y}$ and $\mathbf{Q}\mathbf{y}$.

Solution:

1. $\mathbf{Q}$ is orthogonal iff $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_3$. Write $\mathbf{Q} = \frac{1}{3}\mathbf{M}$ with

$$\mathbf{M} = \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}.$$

The columns of $\mathbf{M}$ are

$$\mathbf{m}_1 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, \quad \mathbf{m}_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \quad \mathbf{m}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Their dot products satisfy

$$\mathbf{m}_i^\top \mathbf{m}_i = 9 \quad (i = 1, 2, 3), \quad \mathbf{m}_i^\top \mathbf{m}_j = 0 \quad (i \neq j),$$

hence $\mathbf{M}^\top \mathbf{M} = 9\mathbf{I}_3$ and

$$\mathbf{Q}^\top \mathbf{Q} = \frac{1}{9} \mathbf{M}^\top \mathbf{M} = \mathbf{I}_3.$$

2. Solve $\mathbf{Q}\mathbf{v} = \mathbf{v}$, i.e. $(\mathbf{Q} - \mathbf{I})\mathbf{v} = \mathbf{0}$. Since $\mathbf{Q} = \frac{1}{3}\mathbf{M}$, this is equivalent to

$$(\mathbf{M} - 3\mathbf{I})\mathbf{v} = \mathbf{0}, \quad \mathbf{M} - 3\mathbf{I} = \begin{bmatrix} -1 & 2 & 1 \\ -2 & -2 & 2 \\ 1 & -2 & -1 \end{bmatrix}.$$

Row-reducing gives $v_2 = 0$ and $v_1 = v_3$, so

$$S = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Take $\mathbf{v}_0 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \in S$. If $\mathbf{y} \perp \mathbf{v}_0$, then

$$\mathbf{v}_0^\top(\mathbf{Q}\mathbf{y}) = (\mathbf{Q}^\top \mathbf{v}_0)^\top \mathbf{y}.$$

Since $\mathbf{v}_0 \in S$, we have $\mathbf{Q}\mathbf{v}_0 = \mathbf{v}_0$, and because $\mathbf{Q}$ is orthogonal $\mathbf{Q}^\top = \mathbf{Q}^{-1}$, hence $\mathbf{Q}^\top \mathbf{v}_0 = \mathbf{v}_0$. Therefore $\mathbf{v}_0^\top(\mathbf{Q}\mathbf{y}) = \mathbf{v}_0^\top \mathbf{y} = 0$, i.e. $\mathbf{Q}\mathbf{y} \perp \mathbf{v}_0$.

Now let $\mathbf{y} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$; note $\mathbf{v}_0^\top \mathbf{y} = 2 + (-2) = 0$, so $\mathbf{y} \perp \mathbf{v}_0$. Compute

$$\mathbf{Q}\mathbf{y} = \frac{1}{3} \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 4/3 \\ -7/3 \\ -4/3 \end{bmatrix}.$$

Then

$$\mathbf{y}^\top(\mathbf{Q}\mathbf{y}) = \begin{bmatrix} 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 4/3 \\ -7/3 \\ -4/3 \end{bmatrix} = 3, \quad \|\mathbf{y}\|^2 = 9, \quad \|\mathbf{Q}\mathbf{y}\|^2 = \|\mathbf{y}\|^2 = 9.$$

Hence the angle θ between $\mathbf{y}$ and $\mathbf{Q}\mathbf{y}$ satisfies

$$\cos \theta = \frac{3}{3 \cdot 3} = \frac{1}{3}, \quad \theta = \arccos\left(\frac{1}{3}\right).$$

Question 3 (15 points)

Work in $\mathbb{R}^3$ with the standard dot product. For $a, b \in \mathbb{R}$ consider

$$\mathbf{P}(a, b) = \begin{bmatrix} \frac{5}{6} & \frac{1}{3} & a \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ b & \frac{1}{3} & \frac{5}{6} \end{bmatrix}.$$

1. (5 points) Find all values for a and b such that $\mathbf{P}(a, b)$ is the matrix of an orthogonal projection.
2. (5 points) For the values found in (1), let $U = \text{Im}(\mathbf{P}(a, b))$. Find a basis for U and a nonzero $\boldsymbol{\omega} \in \text{Ker}(\mathbf{P}(a, b))$, and show that $\boldsymbol{\omega} \perp \mathbf{z}$ for all $\mathbf{z} \in U$.
3. (5 points) For $\mathbf{x} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$, compute $\eta_U(\mathbf{x})$ and $d(\mathbf{x}, U)$.

Solution:

1. A matrix is the orthogonal projection matrix (w.r.t. the standard dot product) iff it is symmetric and idempotent:

$$\mathbf{P}(a, b)^\top = \mathbf{P}(a, b), \quad \mathbf{P}(a, b)^2 = \mathbf{P}(a, b).$$

Symmetry forces the $(1, 3)$ and $(3, 1)$ entries to match, so $a = b$. Write $a = b = \lambda$, so

$$\mathbf{P}(\lambda) = \begin{bmatrix} \frac{5}{6} & \frac{1}{3} & \lambda \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \lambda & \frac{1}{3} & \frac{5}{6} \end{bmatrix}.$$

Since $\mathbf{P}(\lambda)$ is symmetric, it remains to impose $\mathbf{P}(\lambda)^2 = \mathbf{P}(\lambda)$. Compute the $(1, 1)$ -entry of $\mathbf{P}(\lambda)^2$:

$$\left(\frac{5}{6}\right)^2 + \left(\frac{1}{3}\right)^2 + \lambda^2 = \frac{25}{36} + \frac{1}{9} + \lambda^2 = \frac{29}{36} + \lambda^2.$$

Setting this equal to $\frac{5}{6} = \frac{30}{36}$ gives $\lambda^2 = \frac{1}{36}$, so $\lambda = \pm\frac{1}{6}$.

Now compute the $(1, 3)$ -entry of $\mathbf{P}(\lambda)^2$:

$$\frac{5}{6}\lambda + \frac{1}{3} \cdot \frac{1}{3} + \lambda \cdot \frac{5}{6} = \frac{5\lambda}{3} + \frac{1}{9}.$$

Setting this equal to λ yields

$$\frac{5\lambda}{3} + \frac{1}{9} = \lambda \Rightarrow \frac{2\lambda}{3} = -\frac{1}{9} \Rightarrow \lambda = -\frac{1}{6}.$$

Therefore the (unique) choice is

$$a = b = -\frac{1}{6}.$$

2. For $a = b = -\frac{1}{6}$,

$$\mathbf{P} = \begin{bmatrix} \frac{5}{6} & \frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix}.$$

A basis for $U = \text{Im}(\mathbf{P})$ can be obtained from linearly independent columns, e.g.

$$U = \text{span} \left\{ \begin{bmatrix} \frac{5}{6} \\ \frac{1}{3} \\ -\frac{1}{6} \end{bmatrix}, \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix} \right\}.$$

One nonzero vector in $\text{Ker}(\mathbf{P})$ is

$$\boldsymbol{\omega} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad \mathbf{P}\boldsymbol{\omega} = \mathbf{0}.$$

For any $\mathbf{z} \in U$ we can write $\mathbf{z} = \mathbf{P}\mathbf{y}$, hence (using $\mathbf{P}^\top = \mathbf{P}$)

$$\boldsymbol{\omega}^\top \mathbf{z} = \boldsymbol{\omega}^\top \mathbf{P}\mathbf{y} = (\mathbf{P}^\top \boldsymbol{\omega})^\top \mathbf{y} = (\mathbf{P}\boldsymbol{\omega})^\top \mathbf{y} = \mathbf{0}^\top \mathbf{y} = 0,$$

so $\boldsymbol{\omega} \perp \mathbf{z}$ for all $\mathbf{z} \in U$.

3. Since $\boldsymbol{\omega} \perp U$, the projection onto U can be written as

$$\eta_U(\mathbf{x}) = \mathbf{x} - \frac{\boldsymbol{\omega}^\top \mathbf{x}}{\boldsymbol{\omega}^\top \boldsymbol{\omega}} \boldsymbol{\omega}.$$

Compute

$$\boldsymbol{\omega}^\top \mathbf{x} = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} = 5, \quad \boldsymbol{\omega}^\top \boldsymbol{\omega} = 1^2 + (-2)^2 + 1^2 = 6.$$

Therefore

$$\eta_U(\mathbf{x}) = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \frac{5}{6} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{7}{6} \\ \frac{2}{3} \\ \frac{13}{6} \end{bmatrix}.$$

Also,

$$d(\mathbf{x}, U) = \|\mathbf{x} - \eta_U(\mathbf{x})\| = \left\| \frac{5}{6} \boldsymbol{\omega} \right\| = \frac{5}{6} \sqrt{6} = \frac{5}{\sqrt{6}}.$$

Question 4 (15 points)

Work in $\mathbb{R}^4$ with the standard dot product. Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad U = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\}.$$

1. (5 points) Find a basis for $U^\perp$.

2. (6 points) Let $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 3 \end{bmatrix}$. Decompose $\mathbf{x} = \mathbf{p} + \mathbf{q}$ with $\mathbf{p} \in U$ and $\mathbf{q} \in U^\perp$.

3. (4 points) Express $d(\mathbf{x}, U)$ in terms of $\mathbf{q}$.

Solution:

1. Let $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} \in U^\perp$. Then $\mathbf{u}_1^\top \mathbf{v} = 0$ and $\mathbf{u}_2^\top \mathbf{v} = 0$, i.e.

$$v_1 + v_2 = 0, \quad v_2 + v_3 = 0.$$

Hence $v_2 = -v_1$, $v_3 = v_1$, and v_4 is free. Therefore

$$U^\perp = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

2. Use the basis of U and $U^\perp$ from (1). Write

$$\mathbf{x} = a\mathbf{u}_1 + b\mathbf{u}_2 + c \begin{bmatrix} -1 \\ 1 \\ -1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Expanding coordinates:

$$a \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} -1 \\ 1 \\ -1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 3 \end{bmatrix}.$$

This gives the system

$$a - c = 2, \quad a + b + c = 1, \quad b - c = 0, \quad d = 3.$$

From $b = c$, the second equation becomes $a + 2c = 1$. Together with $a - c = 2$:

$$(a + 2c) - (a - c) = 1 - 2 \Rightarrow 3c = -1 \Rightarrow c = -\frac{1}{3},$$

so $b = -\frac{1}{3}$ and $a = 2 + c = \frac{5}{3}$, and $d = 3$.

Therefore

$$\mathbf{p} = a\mathbf{u}_1 + b\mathbf{u}_2 = \frac{5}{3} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{4}{3} \\ \frac{1}{3} \\ 0 \end{bmatrix},$$

and

$$\mathbf{q} = c \begin{bmatrix} -1 \\ 1 \\ -1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = -\frac{1}{3} \begin{bmatrix} -1 \\ 1 \\ -1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ -\frac{1}{3} \\ \frac{1}{3} \\ 3 \end{bmatrix}.$$

3. Since $\mathbf{p} \in U$ and $\mathbf{q} \in U^\perp$, we have $\mathbf{x} - \mathbf{p} = \mathbf{q} \perp U$. Hence $\mathbf{p}$ is the orthogonal projection of $\mathbf{x}$ onto U , and therefore

$$d(\mathbf{x}, U) = \|\mathbf{x} - \mathbf{p}\| = \|\mathbf{q}\|.$$

Question 5 (20 points)

Work in $\mathbb{R}^4$ with the standard dot product. Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 4 \end{bmatrix}, \quad L = \text{span}\{\mathbf{u}\}.$$

1. (8 points) Compute the orthogonal projection $\eta_L(\mathbf{x})$.
2. (6 points) Compute the distance $d(\mathbf{x}, \eta_L(\mathbf{x}))$.
3. (6 points) Let $\mathbf{P}_\eta$ be the matrix of the orthogonal projection onto L . Compute $\mathbf{P}_\eta$ and show that $\mathbf{P}_\eta^\top = \mathbf{P}_\eta$ and $\mathbf{P}_\eta^2 = \mathbf{P}_\eta$.

Solution:

1. For a one-dimensional subspace $L = \text{span}\{\mathbf{u}\}$,

$$\eta_L(\mathbf{x}) = \frac{\mathbf{u}^\top \mathbf{x}}{\mathbf{u}^\top \mathbf{u}} \mathbf{u}.$$

Compute

$$\mathbf{u}^\top \mathbf{x} = 1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 3 + 0 \cdot 4 = -3, \quad \mathbf{u}^\top \mathbf{u} = 1^2 + 2^2 + (-1)^2 + 0^2 = 6.$$

Hence

$$\eta_L(\mathbf{x}) = \frac{-3}{6} \mathbf{u} = -\frac{1}{2} \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/2 \\ -1 \\ 1/2 \\ 0 \end{bmatrix}.$$

2. The distance is $\|\mathbf{x} - \eta_L(\mathbf{x})\|$. Compute the error:

$$\mathbf{x} - \eta_L(\mathbf{x}) = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 4 \end{bmatrix} - \begin{bmatrix} -1/2 \\ -1 \\ 1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 5/2 \\ 0 \\ 5/2 \\ 4 \end{bmatrix}.$$

Therefore

$$d(\mathbf{x}, \eta_L(\mathbf{x})) = \left\| \begin{bmatrix} 5/2 \\ 0 \\ 5/2 \\ 4 \end{bmatrix} \right\| = \sqrt{\frac{25}{4} + \frac{25}{4} + 16} = \sqrt{\frac{57}{2}} = \frac{\sqrt{114}}{2}.$$

3. The projection matrix onto $L = \text{span}\{\mathbf{u}\}$ is

$$\mathbf{P}_\eta = \frac{1}{\mathbf{u}^\top \mathbf{u}} \mathbf{u} \mathbf{u}^\top = \frac{1}{6} \mathbf{u} \mathbf{u}^\top = \frac{1}{6} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -2 & 0 \\ -1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Symmetry is immediate since $(\mathbf{u} \mathbf{u}^\top)^\top = \mathbf{u} \mathbf{u}^\top$, so $\mathbf{P}_\eta^\top = \mathbf{P}_\eta$. Also,

$$\mathbf{P}_\eta^2 = \left(\frac{1}{\mathbf{u}^\top \mathbf{u}} \mathbf{u} \mathbf{u}^\top \right) \left(\frac{1}{\mathbf{u}^\top \mathbf{u}} \mathbf{u} \mathbf{u}^\top \right) = \frac{1}{(\mathbf{u}^\top \mathbf{u})^2} \mathbf{u} (\mathbf{u}^\top \mathbf{u}) \mathbf{u}^\top = \frac{1}{\mathbf{u}^\top \mathbf{u}} \mathbf{u} \mathbf{u}^\top = \mathbf{P}_\eta.$$

Question 6 (20 points)

Work in $\mathbb{R}^3$ with the standard dot product. Let

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad U = \text{span}\{\mathbf{b}_1, \mathbf{b}_2\}.$$

1. (10 points) Find the orthogonal projection $\eta_U(\mathbf{x})$.
2. (4 points) Let decompose $\mathbf{x} = \mathbf{q} + \mathbf{p}$ where $\mathbf{p} \in U$. Prove that $\mathbf{q} \perp \mathbf{b}_1$ and $\mathbf{q} \perp \mathbf{b}_2$.
3. (6 points) Compute the projection matrix $\mathbf{P}_\eta$, and verify that $\mathbf{P}_\eta \mathbf{x} = \eta_U(\mathbf{x})$ and $\mathbf{P}_\eta^\top = \mathbf{P}_\eta$.

Solution:

1. Write $\eta_U(\mathbf{x}) = a\mathbf{b}_1 + b\mathbf{b}_2$ and let

$$\mathbf{r} := \mathbf{x} - \eta_U(\mathbf{x}) = \mathbf{x} - a\mathbf{b}_1 - b\mathbf{b}_2.$$

Since $\mathbf{r} \perp U$, we require $\mathbf{b}_1^\top \mathbf{r} = 0$ and $\mathbf{b}_2^\top \mathbf{r} = 0$:

$$\mathbf{b}_1^\top \mathbf{x} - a\mathbf{b}_1^\top \mathbf{b}_1 - b\mathbf{b}_1^\top \mathbf{b}_2 = 0, \quad \mathbf{b}_2^\top \mathbf{x} - a\mathbf{b}_2^\top \mathbf{b}_1 - b\mathbf{b}_2^\top \mathbf{b}_2 = 0.$$

Compute

$$\mathbf{b}_1^\top \mathbf{b}_1 = 5, \quad \mathbf{b}_2^\top \mathbf{b}_2 = 3, \quad \mathbf{b}_1^\top \mathbf{b}_2 = 1,$$

and

$$\mathbf{b}_1^\top \mathbf{x} = 4, \quad \mathbf{b}_2^\top \mathbf{x} = -1.$$

Hence

$$5a + b = 4, \quad a + 3b = -1.$$

Solving gives $b = -\frac{9}{14}$ and $a = \frac{13}{14}$. Therefore

$$\eta_U(\mathbf{x}) = \frac{13}{14}\mathbf{b}_1 - \frac{9}{14}\mathbf{b}_2 = \begin{bmatrix} \frac{11}{7} \\ \frac{17}{14} \\ \frac{9}{14} \end{bmatrix}.$$

2. Let $\mathbf{p} = \eta_U(\mathbf{x})$ from part (1) and $\mathbf{q} = \mathbf{x} - \mathbf{p}$. From part (1),

$$\mathbf{p} = \eta_U(\mathbf{x}) = \begin{bmatrix} \frac{11}{7} \\ \frac{17}{14} \\ \frac{9}{14} \end{bmatrix}, \quad \mathbf{q} = \mathbf{x} - \mathbf{p} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{11}{7} \\ \frac{17}{14} \\ \frac{9}{14} \end{bmatrix} = \begin{bmatrix} \frac{3}{7} \\ -\frac{3}{14} \\ \frac{9}{14} \end{bmatrix}.$$

Then

$$\mathbf{b}_1^\top \mathbf{q} = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} \frac{3}{7} \\ -\frac{3}{14} \\ \frac{9}{14} \end{bmatrix} = \frac{3}{7} - \frac{3}{7} = 0,$$

and

$$\mathbf{b}_2^\top \mathbf{q} = \begin{bmatrix} -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{3}{7} \\ -\frac{3}{14} \\ \frac{9}{14} \end{bmatrix} = -\frac{3}{7} - \frac{3}{14} + \frac{9}{14} = 0.$$

Hence $\mathbf{q} \perp \mathbf{b}_1$ and $\mathbf{q} \perp \mathbf{b}_2$.

3. Here

$$\mathbf{B} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B}^\top \mathbf{B} = \begin{bmatrix} 5 & 1 \\ 1 & 3 \end{bmatrix}, \quad (\mathbf{B}^\top \mathbf{B})^{-1} = \frac{1}{14} \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix}.$$

Therefore

$$\mathbf{P}_\eta = \mathbf{B}(\mathbf{B}^\top \mathbf{B})^{-1} \mathbf{B}^\top = \begin{bmatrix} 5/7 & 1/7 & -3/7 \\ 1/7 & 13/14 & 3/14 \\ -3/7 & 3/14 & 5/14 \end{bmatrix}.$$

This matrix is symmetric, so $\mathbf{P}_\eta^\top = \mathbf{P}_\eta$. Also,

$$\mathbf{P}_\eta \mathbf{x} = \begin{bmatrix} 5/7 & 1/7 & -3/7 \\ 1/7 & 13/14 & 3/14 \\ -3/7 & 3/14 & 5/14 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{11}{7} \\ \frac{17}{14} \\ -\frac{9}{14} \end{bmatrix} = \eta_U(\mathbf{x}).$$

Bonus Question (10 points)

Let $\mathbf{P} \in \mathbb{R}^{n \times n}$ satisfy $\mathbf{P}^\top = \mathbf{P}$ and $\mathbf{P}^2 = \mathbf{P}$ (under standard dot product).

1. (5 points) Show that $\text{Ker}(\mathbf{P}) = \text{Im}(\mathbf{P})^\perp$.
2. (5 points) Prove that $\mathbf{P}$ is the orthogonal projection matrix onto $\text{Im}(\mathbf{P})$.

Solution:

1. We prove both inclusions.

First, $\text{Ker}(\mathbf{P}) \subseteq \text{Im}(\mathbf{P})^\perp$. Let $\mathbf{k} \in \text{Ker}(\mathbf{P})$, so $\mathbf{P}\mathbf{k} = \mathbf{0}$. Take any $\mathbf{y} \in \text{Im}(\mathbf{P})$, so $\mathbf{y} = \mathbf{P}\mathbf{z}$ for some $\mathbf{z}$. Then, using $\mathbf{P}^\top = \mathbf{P}$,

$$\langle \mathbf{k}, \mathbf{y} \rangle = \mathbf{k}^\top (\mathbf{P}\mathbf{z}) = (\mathbf{P}^\top \mathbf{k})^\top \mathbf{z} = (\mathbf{P}\mathbf{k})^\top \mathbf{z} = \mathbf{0}^\top \mathbf{z} = 0.$$

Hence $\mathbf{k} \perp \text{Im}(\mathbf{P})$, so $\mathbf{k} \in \text{Im}(\mathbf{P})^\perp$.

Second, $\text{Im}(\mathbf{P})^\perp \subseteq \text{Ker}(\mathbf{P})$. Let $\mathbf{k} \in \text{Im}(\mathbf{P})^\perp$. Then $\mathbf{P}\mathbf{k} \in \text{Im}(\mathbf{P})$, so $\mathbf{k} \perp \mathbf{P}\mathbf{k}$ and

$$0 = \langle \mathbf{k}, \mathbf{P}\mathbf{k} \rangle = \mathbf{k}^\top \mathbf{P}\mathbf{k}.$$

Using $\mathbf{P}^\top = \mathbf{P}$, we have

$$\mathbf{k}^\top \mathbf{P}\mathbf{k} = (\mathbf{P}\mathbf{k})^\top \mathbf{k}.$$

Also, since $\mathbf{P}^2 = \mathbf{P}$,

$$\|\mathbf{P}\mathbf{k}\|^2 = (\mathbf{P}\mathbf{k})^\top (\mathbf{P}\mathbf{k}) = (\mathbf{P}\mathbf{k})^\top \mathbf{P}\mathbf{k} = \mathbf{k}^\top \mathbf{P}^\top \mathbf{P}\mathbf{k} = \mathbf{k}^\top \mathbf{P}^2 \mathbf{k} = \mathbf{k}^\top \mathbf{P}\mathbf{k} = 0.$$

Thus $\mathbf{P}\mathbf{k} = \mathbf{0}$, i.e. $\mathbf{k} \in \text{Ker}(\mathbf{P})$.

Therefore $\text{Ker}(\mathbf{P}) = \text{Im}(\mathbf{P})^\perp$.

2. Let $U = \text{Im}(\mathbf{P})$. For any $\mathbf{x} \in \mathbb{R}^n$, we have the decomposition

$$\mathbf{x} = \mathbf{P}\mathbf{x} + (\mathbf{I} - \mathbf{P})\mathbf{x},$$

with $\mathbf{P}\mathbf{x} \in U$ by definition. Moreover,

$$\mathbf{P}((\mathbf{I} - \mathbf{P})\mathbf{x}) = \mathbf{P}\mathbf{x} - \mathbf{P}^2\mathbf{x} = \mathbf{P}\mathbf{x} - \mathbf{P}\mathbf{x} = \mathbf{0},$$

so $(\mathbf{I} - \mathbf{P})\mathbf{x} \in \text{Ker}(\mathbf{P})$. By part (1), $\text{Ker}(\mathbf{P}) = U^\perp$, hence

$$(\mathbf{I} - \mathbf{P})\mathbf{x} \perp U.$$

Thus $\mathbf{x} - \mathbf{P}\mathbf{x} \perp U$, i.e. $\mathbf{P}\mathbf{x}$ is the orthogonal projection of $\mathbf{x}$ onto U .

Equivalently, for any $\mathbf{y} \in U$,

$$\mathbf{x} - \mathbf{y} = (\mathbf{x} - \mathbf{P}\mathbf{x}) + (\mathbf{P}\mathbf{x} - \mathbf{y}),$$

where $\mathbf{x} - \mathbf{P}\mathbf{x} \in U^\perp$ and $\mathbf{P}\mathbf{x} - \mathbf{y} \in U$, so the two terms are orthogonal. Hence

$$\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x} - \mathbf{P}\mathbf{x}\|^2 + \|\mathbf{P}\mathbf{x} - \mathbf{y}\|^2 \geq \|\mathbf{x} - \mathbf{P}\mathbf{x}\|^2,$$

with equality iff $\mathbf{y} = \mathbf{P}\mathbf{x}$. Therefore $\mathbf{P}\mathbf{x}$ is the unique closest vector in U to $\mathbf{x}$, i.e. the orthogonal projector onto $U = \text{Im}(\mathbf{P})$.